Lecture 15 (10/19)

More examples of multiple predictor regression

Last Week:

- In observational data (unlike a randomized trial), other factors can affect both X and Y. These are called confounding variables.
- Multiple-predictor regression $Y = \beta_0 + \sum_k \beta_k X_k + \epsilon$: β_k is the difference in average Y per unit of X_k controlling for (meaning holding fixed) the other predictors.
- Omitted variable bias is the difference between the single-predictor slope β_1' and the multiple-regression slope β_1. We also saw adding a predictor never decreases ρ^2 (or R^2).
- Examples: test scores and mother info, violent crime, and baseball (the second two were examples of Simpson's paradox).

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Today:

1) Omitted variables in AR processes
2) Incumbency advantage in congressional elections; categorical predictors

Recall the AR(1) process

$$Y_t = \rho Y_{t-1} + \epsilon_t.$$

Assume the process is stationary (all Y_t have the same marginal distribution). Suppose data is generated from an AR(1) process, fit to the model

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \epsilon'_t.$$

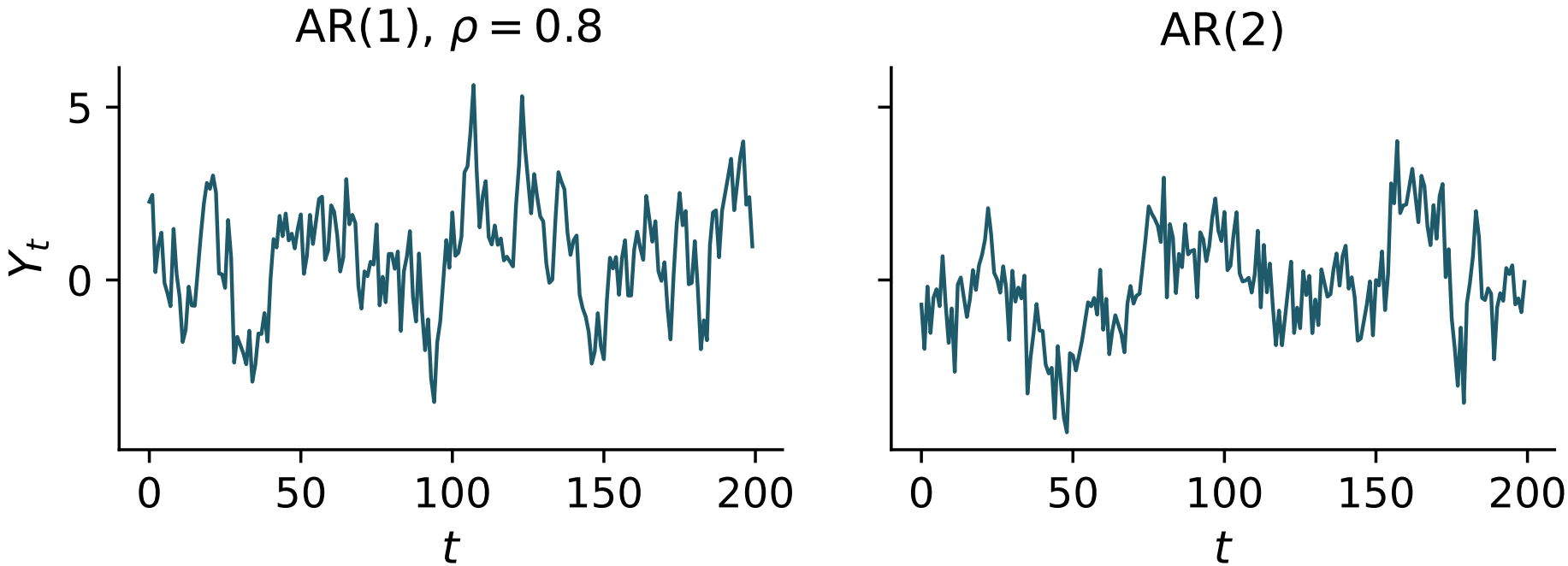
Question

Assuming a large amount of data, what will $\hat{\beta}_1$ and $\hat{\beta}_2$ be? Derive the regression coefficient when Y_{t-1} is omitted from the model using OVB. Then generalize this to obtain the regression coefficient between Y_{t-k} and Y_t.

An AR(2) process

Now let Y_t depend on *two* previous values:

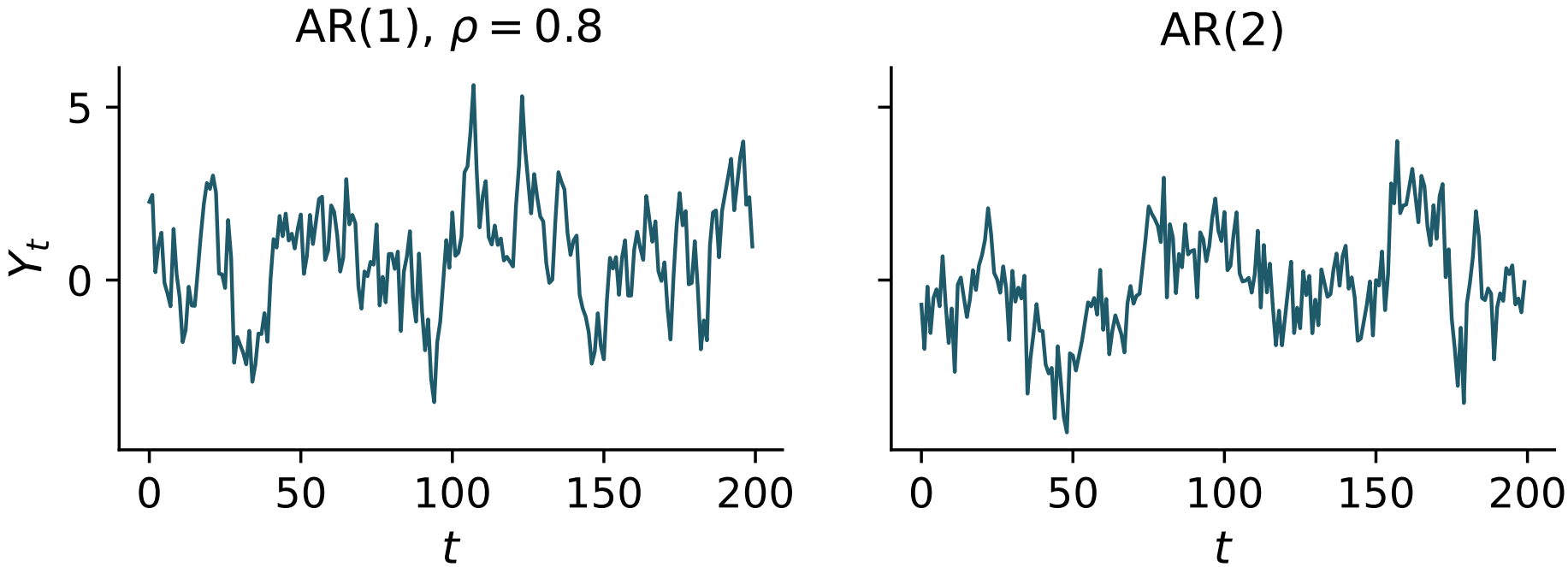
$$Y_t = 0.5\, Y_{t-1} + 0.3\, Y_{t-2} + \epsilon_t, \qquad \epsilon_t \sim \text{Normal}(0, 1).$$



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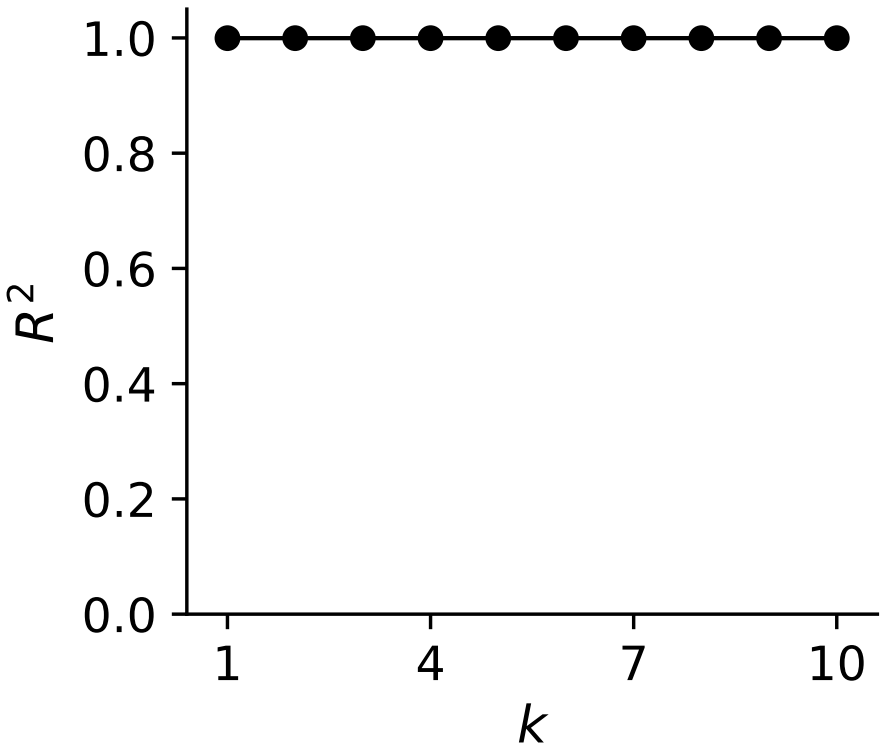


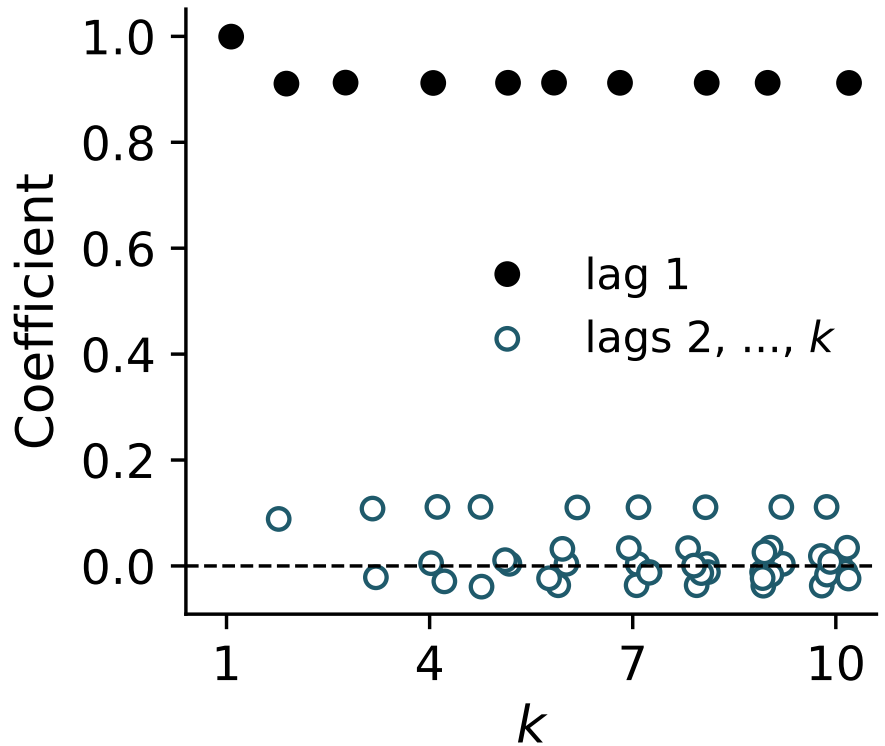
Question

What is the slope of the single-predictor regression of Y_t on Y_{t-1} (with Y_{t-2} excluded)?

Bitcoin: AR(k) models

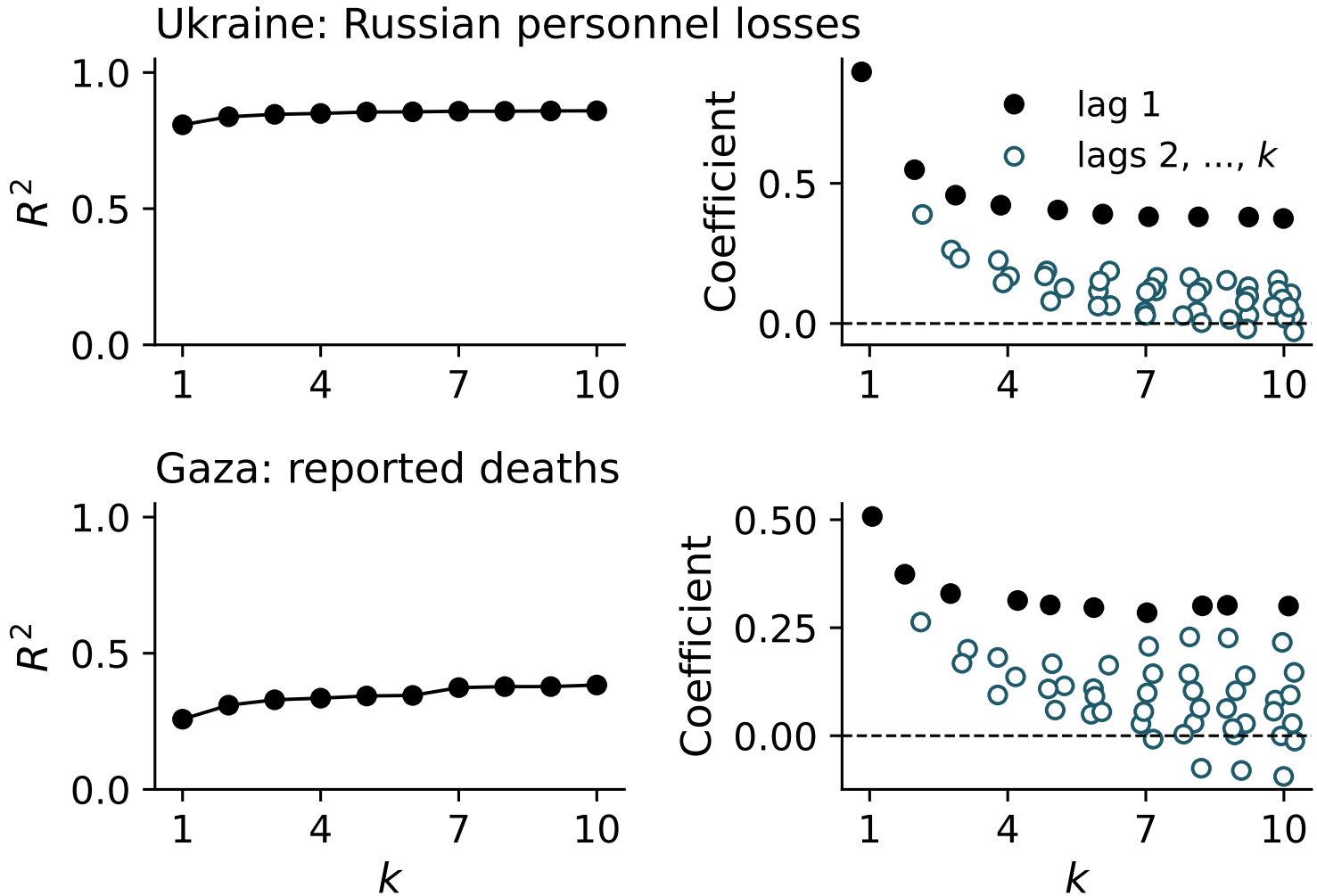
$Y_t = \log_{10}$(daily bitcoin price), regressed on its previous k values $Y_{t-1}, \ldots, Y_{t-k}$ for $k = 1, \ldots, 10$ (the same 4283 days in every fit):





Casualty counts: AR(k) models

Daily counts (changes in the cumulative series), regressed on the previous k days, $k = 1, \ldots, 10$:



Example

How big is the incumbency advantage? US House elections, 435 districts (from *Regression and Other Stories*):

- Y (`vote`): Democratic share of the two-party vote in 1988, as a fraction from 0 to 1 (0.55 means 55%);
- X_{past} (`past_vote`): Democratic share in 1986, on the same 0 to 1 scale;
- X_{inc} (`inc`): $+1$ if a Democrat is the incumbent, -1 if a Republican is, 0 for an open seat.

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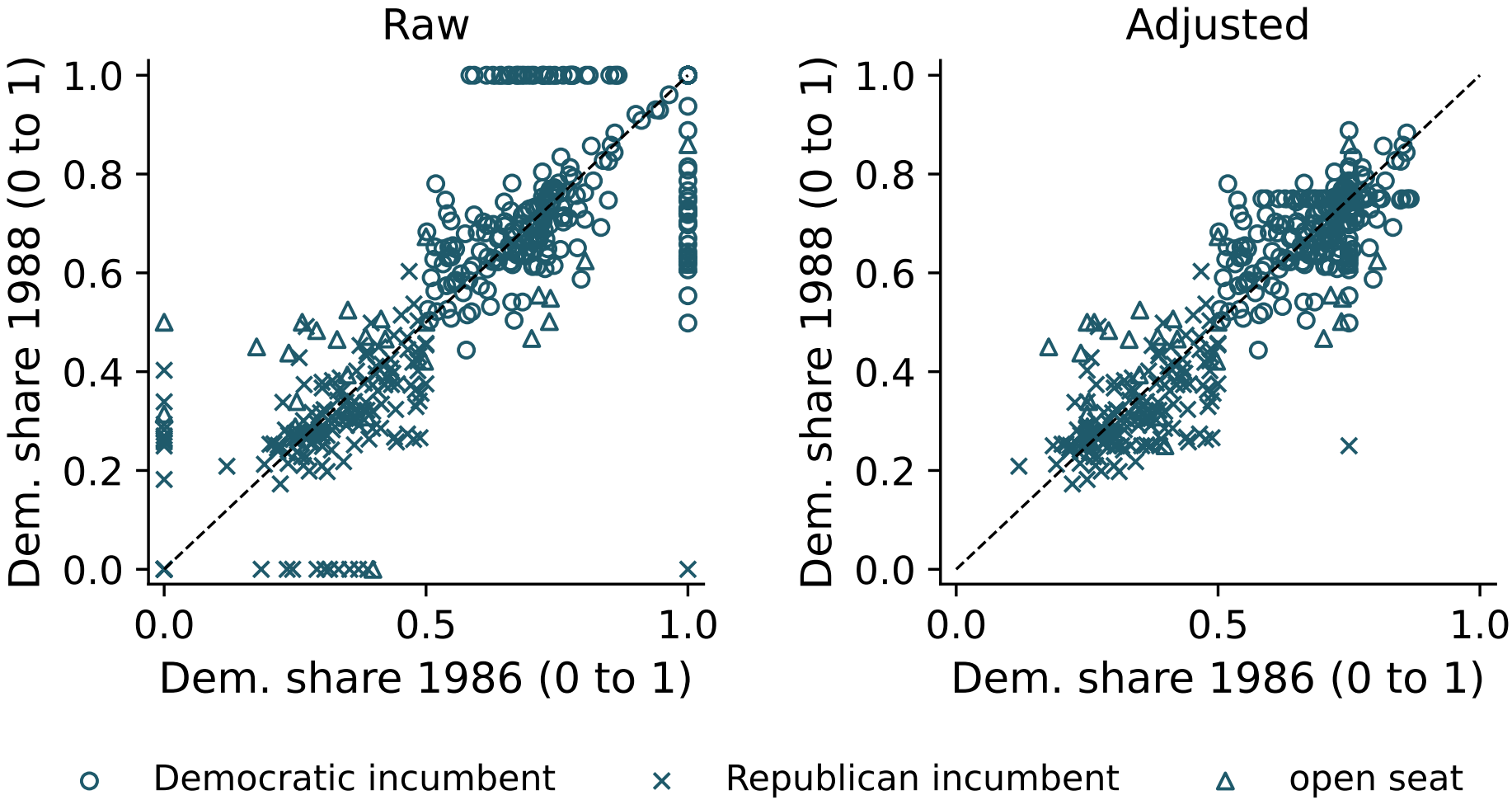
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Question

What is at least one limitation of this model? (You can assume this is the only data we have, so the limitation is about the assumptions)

1988 vs. 1986 vote



Each point is one district. The dashed line is $y = x$. Adjusted: uncontested races set to 0.25 or 0.75.

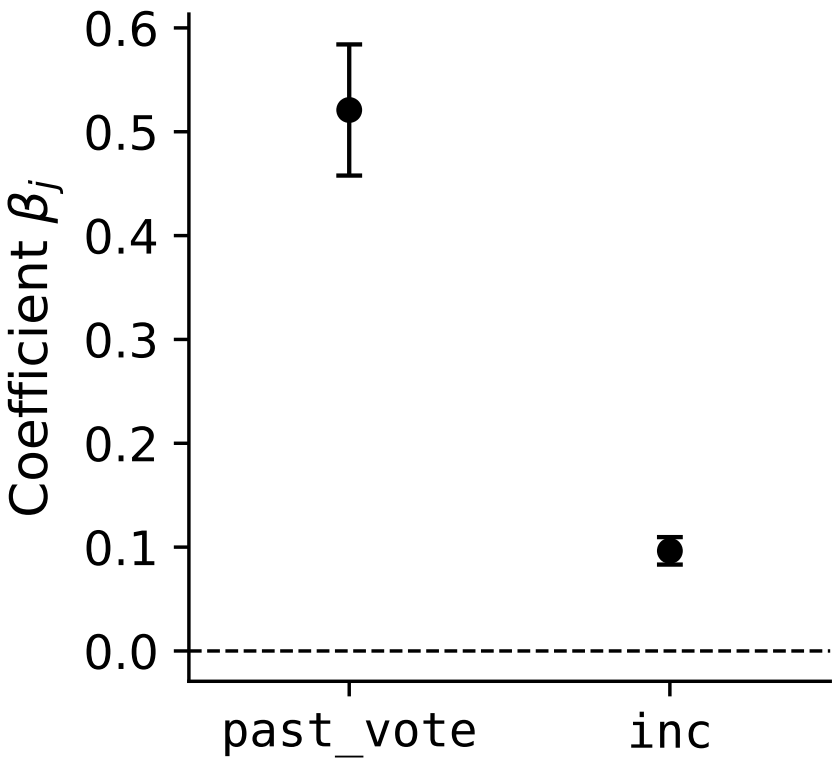
Dep. Variable:			vote	R-squared:		0.819
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.5063	0.004	121.695	0.000	0.498	0.514
inc	0.1906	0.004	44.210	0.000	0.182	0.199
Dep. Variable:			vote	R-squared:		0.887
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.2380	0.017	14.068	0.000	0.205	0.271
past_vote	0.5210	0.032	16.169	0.000	0.458	0.584
inc	0.0964	0.007	14.284	0.000	0.083	0.110

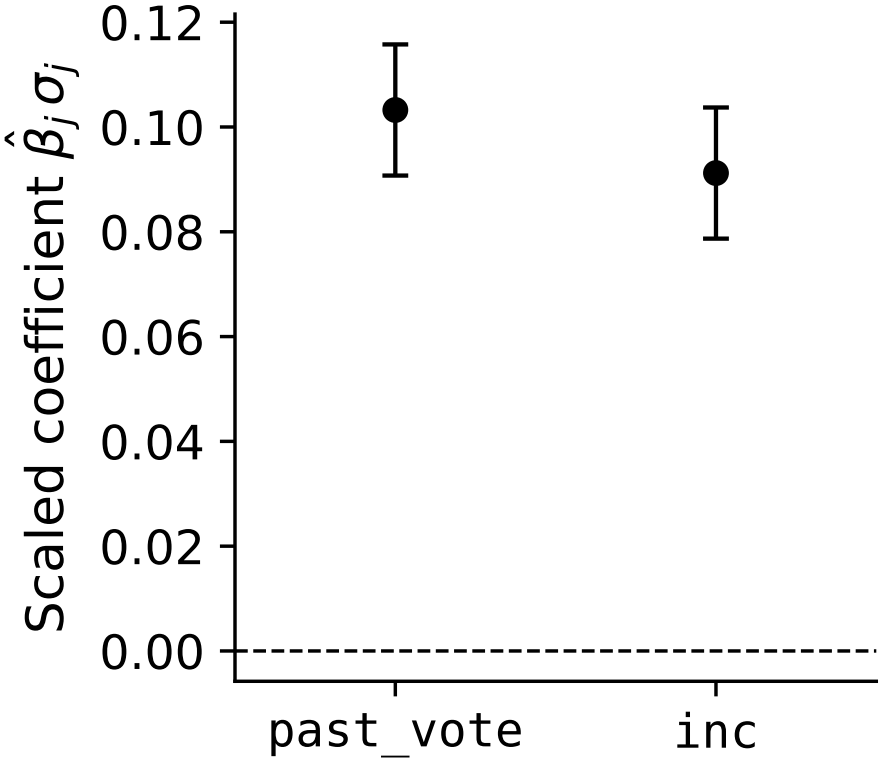
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Question

Interpret the coefficient on `inc` in each model. Approximately what is the regression coefficient of `inc` when `past_vote` is the response variable? Compare the importance of the two predictors. The standard deviations of the predictors are $\sigma_{\text{past}} \approx 0.20$ and $\sigma_{\text{inc}} \approx 0.95$.

Coefficients vs. scaled coefficients





Categorical predictors

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  - X_{dem} (`dem_inc`): Democratic incumbent;
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The model above has a problem, which is seen when we try to answer the following question.

Question

What is the interpretation of β_{dem}?

Incumbency with dummy variables

Solution: leave out one category, the baseline (here, open seats). We compare the linear and categorical models:

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const	0.2530	0.020	12.947	0.000	0.215	0.291
past_vote	0.5270	0.032	16.260	0.000	0.463	0.591
dem_inc	0.0760	0.015	5.089	0.000	0.047	0.105
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Question

Which choice of β_{dem} and β_{rep} gives back the linear model? Compare $\hat{\beta}_{\text{dem}}$ and $\hat{\beta}_{\text{rep}}$ with $\hat{\beta}_{\text{inc}}$. Is the incumbency advantage the same for both parties?